

Shahaf Bassan

PHD CANDIDATE,

INTERESTS: EXPLAINABILITY & MECHANISTIC INTERPRETABILITY, ML VERIFICATION, ML THEORY, ROBUSTNESS, OPTIMIZATION

Hebrew University of Jerusalem

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Education

Hebrew University of Jerusalem

PH.D. IN COMPUTER SCIENCE

• Advisor: Prof. Guy Katz

Jerusalem, Israel

January 2023 - Current

Hebrew University of Jerusalem

M.SC. IN COMPUTER SCIENCE

• Advisor: Prof. Jeff Rosenschein co-advised with Dr. Yossi Adi

Jerusalem, Israel

2021 - 2022

Hebrew University of Jerusalem

B.SC. IN COMPUTER SCIENCE AND PHYSICS

Jerusalem, Israel

2017 - 2020

Publications

Shahaf Bassan*, Yizhak Elboher*, Itamar Hadad*, Jan Kretinsky, Guy Katz. Provably Optimal Compression for LLM Tokenization [*Under submission*]

Idan Refaeli, **Shahaf Bassan*** Yizhak Elboher, Guy Katz. Provable Explanations for Any-Order Neural Additive Models [*Under submission*]

David Boetius*, **Shahaf Bassan***, Guy Katz, Stefan Leue, Tobias Sutter. Verified SHAP: Provable Bounds for Exact Shapley Values of Neural Networks [*ICML, 2026, Seoul*][*Paper*]

Itamar Hadad, Guy Katz, **Shahaf Bassan**. Formal Mechanistic Interpretability: Automated Circuit Discovery with Provable Guarantees [*ICLR, 2026, Rio De Janeiro*][*Paper*]

Shahaf Bassan*, Xuanxiang Huang*, Guy Katz. Unifying Formal Explanations: A Complexity-Theoretic Perspective [*ICLR, 2026, Rio De Janeiro*][*Paper*]

Shahaf Bassan*, Yizhak Elboher*, Tobias Ladner*, Volkan Sahin, Jan Kretinsky, Matthias Althoff, Guy Katz. Provably Explaining Neural Additive Models [*ICLR, 2026, Rio De Janeiro*][*Paper*]

Ryma Boumazouza, Raya Elsaleh, Melanie Ducoffe, **Shahaf Bassan**, Guy Katz. FAME: Formal Abstract Minimal Explanation for Neural Networks [*ICLR, 2026, Rio De Janeiro*][*Paper*]

Shahaf Bassan, Michal Moshkovitz, Guy Katz. Additive Models Explained: A Computational Complexity Approach [*NeurIPS, 2025, San-Diego*][*Paper*]

Reda Marzouk*, **Shahaf Bassan*** (equal contribution), Guy Katz. SHAP Meets Tensor Networks: Provably Tractable Explanations with Parallelism [*NeurIPS, 2025, San Diego*][*Paper*]

Shahaf Bassan, Guy Amir, Meirav Zehavi, Guy Katz. What makes an Ensemble (Un) Interpretable? [*ICML, 2025, Vancouver*][*Paper*]

Shahaf Bassan*, Yizhak Elboher*, Tobias Ladner* (equal contribution), Matthias Althoff, Guy Katz. Explaining, Fast and Slow: Abstraction and Refinement of Provable Explanations [*ICML, 2025, Vancouver*][*Paper*]

Shahaf Bassan, Ron Eliav, Shlomit Gur. Explain Yourself, Briefly! Self-Explaining Neural Networks with Concise Sufficient Reasons [*ICLR, 2025, Singapore*][*Paper*]

Reda Marzouk*, **Shahaf Bassan*** (equal contribution), Guy Katz, Colin de la Higuera. On the Computational Tractability of the (Many) Shapley Values [*AISTATS, 2025, Mai Khao, Thailand*][*Paper*]

Shahaf Bassan, Guy Amir, Guy Katz. Local vs. Global Interpretability: A Computational Complexity Perspective [*ICML, 2024, Vienna, Austria*] (*Spotlight*) [*Paper*]

Ron Eliav, Arie Cattan, Eran Hirsch, **Shahaf Bassan**, Elias Stengel-Eskin, Mohit Bansal, Ido Dagan. CLATTER: Comprehensive Entailment Reasoning for Hallucination Detection [*preprint*, 2025][[Paper](#)]

Shahaf Bassan*, Shlomit Gur*, Sergey Zeltyn*, Konstantinos Mavrogiorgos*, Ron Eliav, Dimosthenis Kyriazis. Self-Explaining Neural Networks for Business Process Monitoring [*ICSBT*, 2025, *Balboa, Spain*, *Best Paper Runnerup*][[Paper](#)]

Guy Amir*, **Shahaf Bassan*** (equal contribution), Guy Katz. Hard to Explain: On the Computational Hardness of In-Distribution Model Interpretation [*ECAI*, 2024, *Santiago, Spain*][[Paper](#)]

Haoze Wu, Omri Isac, Aleksandar Zeljić, Teruhiro Tagomori, Matthew Daggitt, Wen Kokke, Idan Refaeli, Guy Amir, **Shahaf Bassan**, Ori Lahav, Min Wu, Min Zhang, Ekaterina Komendantskaya, Guy Katz, and Clark Barrett. Marabou 2.0: A Versatile Formal Analyzer of Neural Networks [*CAV* 2024, *Montreal, Canada* [[Paper](#)]

Shahaf Bassan*, Guy Amir*, Davide Corsi, Idan Refaeli, Guy Katz. Formally Explaining Neural Networks within Reactive Systems [*FMCAD*, 2023, *Iowa, USA*, *Best Paper Runnerup*][[Paper](#)]

Shahaf Bassan, Guy Katz. Towards Formal XAI: Formally Approximate Minimal Explanations of Neural Networks. [*Tacas*, 2023, *Paris, France*][[Paper](#)]

Shahaf Bassan, Yossi Adi, Jeffrey S. Rosenschein. Symbolic Music Segmentation using Ensemble Temporal Prediction Errors [*Interspeech*, 2022, *Incheon, South Korea*] [[Paper](#)]

Teaching Experience

Data Structures and Algorithms, Teaching Assistant.

2021-2023

2021, 2022, 2023: Chosen for the list of outstanding teachers at the Hebrew University.
2022, 2023: Ranked **first** in student surveys among all computer science and engineering faculty members at the Hebrew University, out of hundreds of teachers and lecturers.

Invited Talks

December 2025 **Invited Keynote Speaker** at the Theory of Explainable ML Workshop, Copenhagen [[Workshop link](#)]

October 2025 **Invited Speaker** at the Theory of Interpretable AI Seminar: “The Computational Complexity of Explaining ML Models” [[Talk](#)]

June 2025 **Invited Speaker** at the IARCS Verification Seminar: “Formal XAI”: Can we formally explain ML models? [[Slides](#)]

March 2025 **Invited Speaker** at the Interpretability Research Group, Tel Aviv: “Explain Yourself, Briefly! Self-Explaining Neural Networks with Concise Sufficient Reasons”

February 2025 **Invited Speaker** at Kings College, London: “Towards Formal Explainability of Neural Networks”

July 2024 **Invited Speaker** at Bosch Research Center for AI, Haifa: “Formal XAI: Formally Explaining the Decisions of ML Models”

September 2023 **Invited Speaker** at IBM Research, Haifa: “Formal XAI: Can we Formally Explain ML Models?”

June 2023 **Keynote Speaker** at Mobilit.AI, Toulouse, France: “Towards Formally Verifying and Explaining Deep Neural Networks” [[Talk](#)]

April 2023 **Invited Speaker** at TADM Workshop, Paris, France: “Formal Verification of Large Language Models”

April 2023 **Invited Speaker** at Live Workshop, Paris, France: “Formal Explainability of Deep Neural Networks”

Professional Experience

- Explainable AI Research Intern**, IBM Research (Summer Internship)
2023 Research Topic: Explainability during training. Two papers published during the internship, one at ICLR (see publications)
2019-2020 **Software Engineer**, Fiverr (Key involvement in building a new major feature: "Fiverr for Business")

Volunteer Experience

- Alumni Forum Co-Leader and Key Contributor**, Tech2Peace
2020-2024 **Tech2Peace** is an NGO with a noble mission to bring Israelis and Palestinians together through technology and entrepreneurship.
2021-Present **Programming Teacher and Lecturer**, Tech2Peace

Honors and Awards

- 2026 **Hebrew University of Jerusalem**, Chosen as 1 of 5 research students at HUJI university-wide, and the only CS & Engineering representative, for the "Journey to Science" research showcase
2026 **ICML**, Outstanding (gold) reviewer award
2025 **ICML**, Outstanding reviewer award
2025 **ICSBT**, Best paper runnerup: *Self-Explaining Neural Networks for Business Process Monitoring*
2024 **FMCAD**, Best paper runnerup: *Formally Explaining Neural Networks within Reactive Systems*
2024 **ICML**, Spotlight paper award: *Local vs. Global Interpretability: A Computational Complexity Perspective*
2024 **Federman Security Center**, Federman Scholarship for Excellent PhD students
2023 **Hebrew University of Jerusalem**, Outstanding teacher award
2023 **Hebrew University of Jerusalem**, Ranked first in student surveys among all computer science and engineering faculty members at the Hebrew University, out of hundreds of teachers and lecturers.
2023 **Federman Security Center**, Federman Scholarship for Excellent PhD students
2022 **Hebrew University of Jerusalem**, Outstanding teacher award
2022 **Hebrew University of Jerusalem**, Ranked first in student surveys among all computer science and engineering faculty members at the Hebrew University, out of hundreds of teachers and lecturers.
2021 **Hebrew University of Jerusalem**, Outstanding teacher award

Community Service

- 2025-Present **Theory of Interpretable AI seminar (an international monthly seminar on the theoretical foundations of interpretability)**, Co-Organizer
2025, 2026 **Selected as an outstanding reviewer**, ICML
2024-Present **ICLR, ICML, NeurIPS, Tacas**, Reviewer